Case Study 6: Password Policy Validation Engine

Assignment Case No. 29 | Level: Beginner

01 PROBLEM STATEMENT

Evaluate sample password strings against minimum length, character-category, repetition, and prohibited-substring rules; explain each failed condition.

Problem interpretation: Check sample passwords against a basic policy and report every rule that a password fails. This is a classroom validation exercise, not a complete security audit.

02 OBJECTIVES

Check minimum password length.

Check uppercase, lowercase, digit, and special-character requirements.

Check for a repeated character sequence and a prohibited substring.

Report all failed rules rather than stopping at the first failure.

03 ASSUMPTIONS

Minimum length is 8 characters.

The password must include uppercase, lowercase, a digit, and a special character.

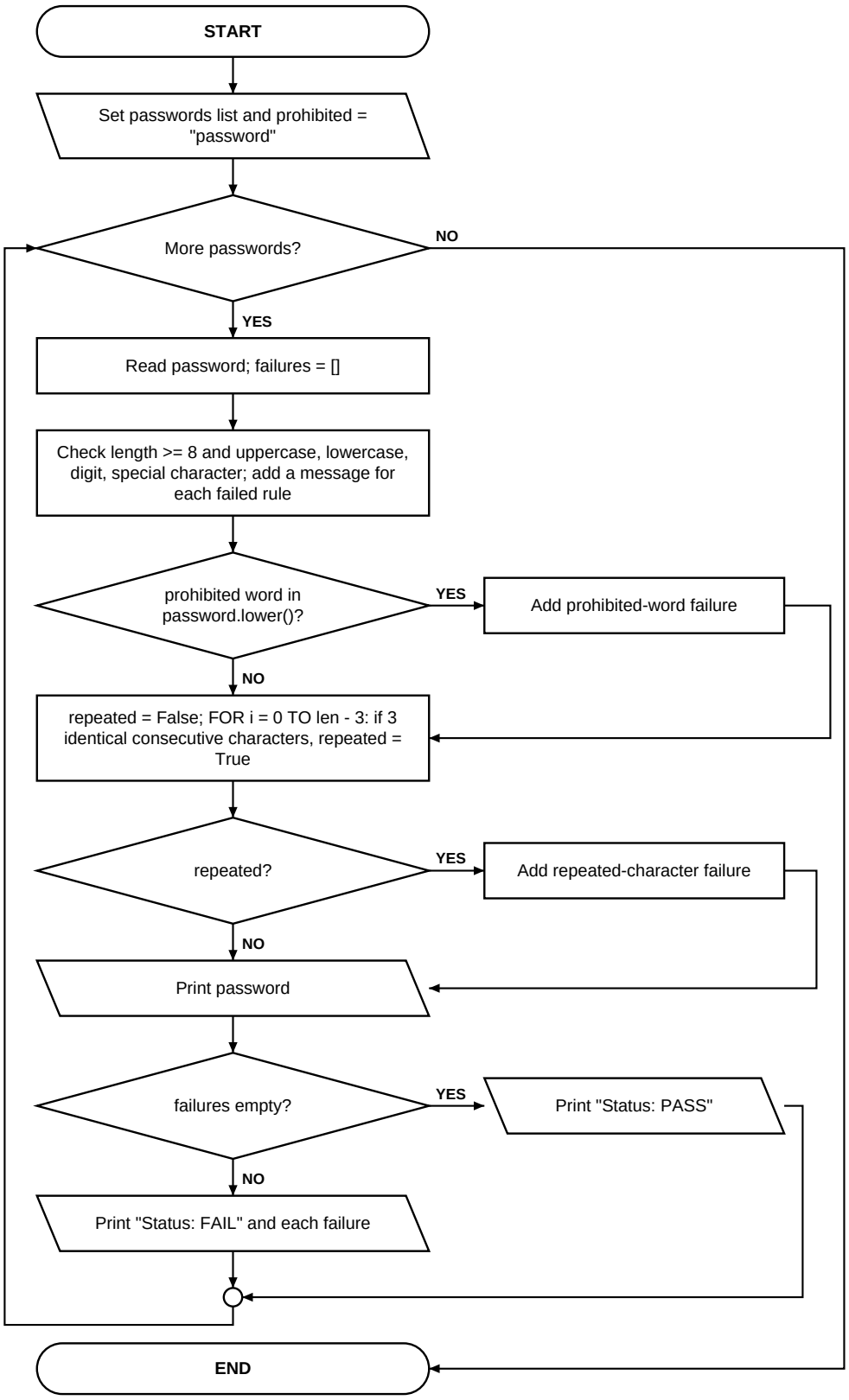
The substring 'password' is prohibited, ignoring letter case.

Three consecutive identical characters are not allowed.

04 ALGORITHM

1. Start.
2. Store sample passwords.
3. For each password, create an empty list of failures.
4. Check length and each required character category.
5. Check the prohibited substring and three repeated consecutive characters.
6. If no failures exist, label the password as passing the sample policy.
7. Otherwise print every failure.
8. Stop.

05 FLOWCHART (TEXT FLOW)



06 PSEUDOCODE

```
BEGIN
    passwords = [Hello@123, passwordA1!, abc111XY!, short]
    prohibited = "password"
    FOR each password IN passwords
        failures = []
        IF length(password) < 8 THEN add length failure
        IF no uppercase letter THEN add uppercase failure
        IF no lowercase letter THEN add lowercase failure
        IF no digit THEN add digit failure
        IF no special character THEN add special-character failure
        IF prohibited IN lowercase(password) THEN add prohibited-word failure
        repeated = False
        FOR i = 0 TO length(password) - 3
            IF password[i] = password[i+1] = password[i+2] THEN repeated = True
        END FOR
        IF repeated THEN add repeated-character failure
        PRINT password
        IF failures is not empty THEN
            PRINT "Status: FAIL"
            FOR each failure: PRINT failure
        ELSE
            PRINT "Status: PASS"
        END IF
    END FOR
END
```

07 PYTHON SOURCE CODE

The complete Python program is provided in the separate file 06_Password_Policy_Validation_Engine_Code.pdf.

08 TEST CASES

INPUT / SCENARIO	CHECK	EXPECTED RESULT
Hello@123	All rules satisfied	PASS
passwordA1!	Contains the prohibited word 'password'	FAIL
abc111XY!	Three repeated consecutive digits	FAIL
short	Too short; no uppercase, digit, or special character	FAIL

09 SAMPLE OUTPUT PREVIEW

```
Password: Hello@123
Status: PASS

Password: passwordA1!
Status: FAIL
- contains prohibited word 'password'

Password: abc111XY!
Status: FAIL
- has 3 identical consecutive characters

Password: short
Status: FAIL
- must contain at least 8 characters
- needs an uppercase letter
- needs a digit
- needs a special character
```

10 CONCLUSION

The program uses string methods, loops, conditions, and a list of error messages to explain which password rules are not met.